Experiment 3 – Optical Signal Processing

Explanation: purpose, principle, procedure and why it matters later

1. The experiment in one sentence

An optocoupler whose output is fed back to its own input becomes a **bistable switch**. A single cell can store one bit (optically addressed memory) or act as a Schmitt trigger, and two coupled cells form an RS flip-flop. Light is used for writing, erasing and reading out.

2. Purpose of the experiment

- Understand how **optical bistability (hysteresis)** arises from a nonlinear element combined with **positive feedback**.
- Measure and interpret **current–voltage (I–U) characteristics**, quasi-statically with a plotter and dynamically with an oscilloscope in X-Y mode.
- Understand the role of every resistor and of the **load line** set by the series resistor R_V.
- Build an optically addressed **memory**, an optical **Schmitt trigger** and an optical **RS flip-flop**, and verify the truth table.
- Compare **simulation (LTspice) and measurement**.

3. Principle – the logic behind it

3.1 Optical bistability

A bistable element has two possible output states for the same input. Which one it shows depends on its history, so it is a **memory and a switch** at the same time. Light beams do not interact in a linear medium, so light cannot directly switch light. Either special nonlinear materials are needed, or a **hybrid optoelectronic** solution: light → photocurrent → electronics → light. This experiment uses the hybrid route, an "artificial optical nonlinearity".

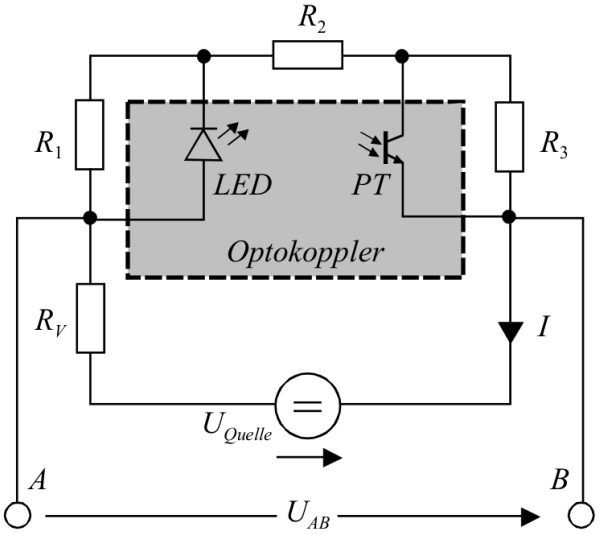


Fig. 1: Feedback optocoupler (from the lab documentation).

3.2 The feedback optocoupler

The current flows through the chain (R_1 ‖ LED) → R_2 → (R_3 ‖ phototransistor). At low voltage the LED is dark, the phototransistor is off and the large resistance R_1+R_2+R_3 limits the current (flat OFF branch). As soon as the voltage across R_1 reaches the LED forward voltage ($I \cdot R_1 \approx U_F$), the LED emits light and the phototransistor starts to conduct. It short-circuits the largest resistor R_3, so the current rises, the LED gets brighter and the transistor conducts even more. This **positive feedback** (loop gain > 1) makes the current jump to the steep ON branch.

When the voltage is reduced, the transistor keeps R_3 short-circuited down to a lower voltage. Switch-on and switch-off happen at different voltages, which is the hysteresis.

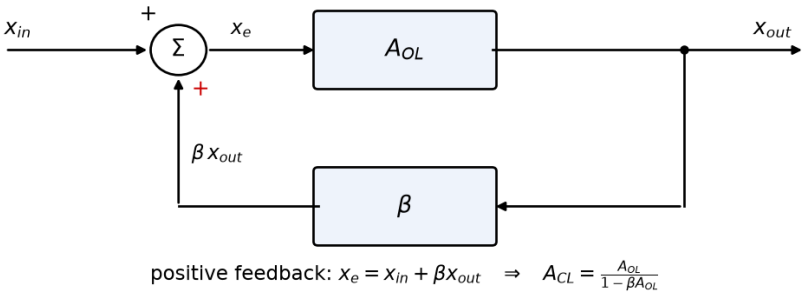


Fig. 2: Positive feedback, $A_{CL} = A_{OL}/(1 - \beta \cdot A_{OL})$. For $\beta \cdot A_{OL} \geq 1$ the system latches.

Rule of thumb for the switch-on voltage: $\mathbf{U_{on} \approx U_F \cdot (R_1+R_2+R_3)/R_1}$. Increasing R_1 lowers both thresholds. Increasing R_2 raises them and narrows the hysteresis. Increasing R_3 raises only U_{on}, so the hysteresis widens.

3.3 S-curve, negative resistance and the load line

Plotted against the current, the characteristic is S-shaped. In the middle part the voltage decreases while the current increases (**negative differential resistance**), and this part is unstable when the voltage is imposed. The series resistor defines a load line $U_{AB} = U_{source} - I \cdot R_V$. With small R_V the circuit jumps (hysteresis). With R_V larger than the negative resistance (≈ 0.82 kΩ in LTspice) the whole S-curve can be traced. A thyristor has the same kind of curve.

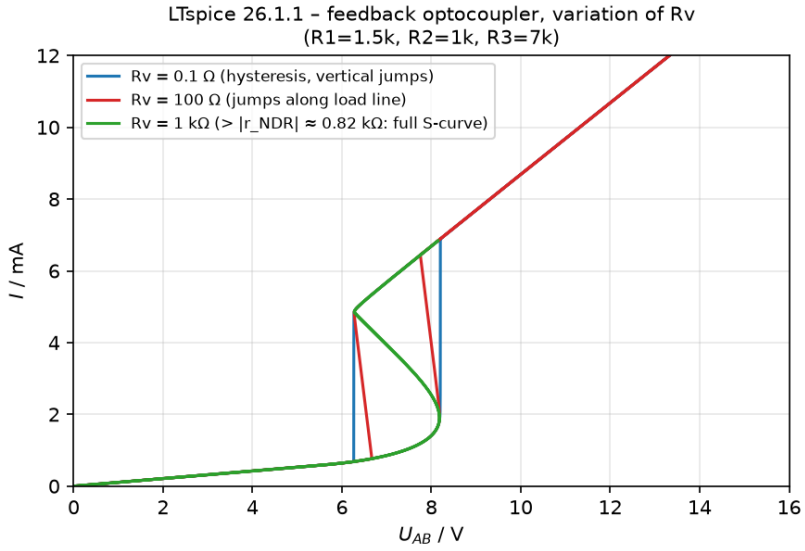


Fig. 3: LTspice – the series resistor R_V turns the hysteresis into the S-curve.

3.4 Three circuits from one cell

- **Memory:** the operating point lies inside the bistable range. Light on a photodiode in parallel to R_3 switches ON (SET), and the state remains after the light is switched off. Light on an LDR in parallel to R_1 switches OFF (RESET).
- **Schmitt trigger:** the operating point lies outside the bistable range in the dark, so the input light itself is the input signal. The output switches at two different input thresholds and needs a continuous input to stay ON.
- **RS flip-flop:** two identical cells share one series resistor R_V. When one cell switches ON, the voltage U_{AB} drops so far that the other cell cannot stay ON, so only one LED lights (Q or $\bar{Q}$).

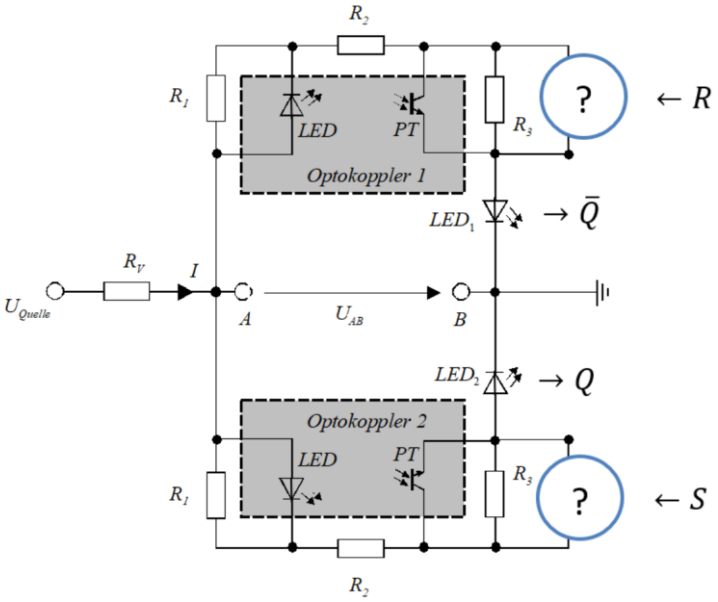


Fig. 4: Two cells coupled to an RS flip-flop (from the lab documentation).

4. Procedure step by step

6.1 Feedback optocoupler

- Quasi-static: DC source + plotter (or function generator at ≈ 0.1 Hz). Dynamic: triangular voltage + oscilloscope in X-Y mode.
- Basic values R_1 = 20 %, R_2 = 50 %, R_3 = 80 %, R_V = 0 %. Change **one** resistor at a time and record U_{on} and U_{off}.
- Expected (LTspice, nominal values): switch-on 8.20 V, switch-off 6.27 V; the trends follow section 3.2.

6.2 Optically addressed memory

- Record the I–U curve, then set the DC voltage between the two jump points.
- Illuminate the photodiode ∥ R_3: the circuit switches ON and stays ON. Illuminate the LDR ∥ R_1: it switches OFF. (LTspice: SET for an input of ≥ 0.3 mA LED current, RESET for $R_{LDR} \lesssim 3$ kΩ.)

6.3 Optical Schmitt trigger

- U_{source} = 5 V; LED_3 is driven by the function generator and illuminates the detector ∥ R_3. Display I_{out} versus I_{in} in X-Y mode.
- Triangle 50 Hz: hysteresis loop. Square 50 Hz: only two points (no intermediate values). Square 2 kHz: the loop is distorted because the phototransistor is slow (µs), giving a dynamic hysteresis.

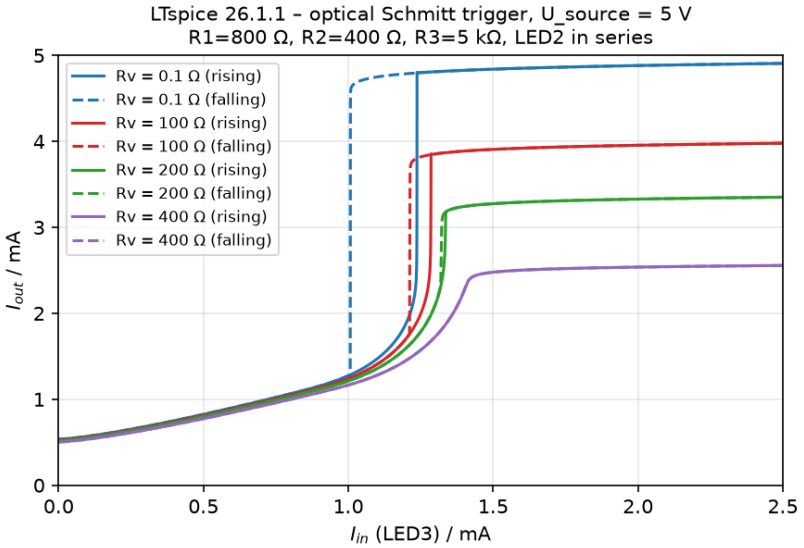


Fig. 5: LTspice – Schmitt trigger (R_1 = 800 Ω, R_2 = 400 Ω, R_3 = 5 kΩ). The hysteresis narrows with increasing R_V.

6.4 Optical RS flip-flop

- R_1 = 20 %, R_2 = 50 %, R_3 = 100 %. Find a suitable R_V with the oscilloscope, then use the DC source. Illuminate S and R alternately and fill in the truth table.

R	S	$\bar{Q}$ (LED_1)	Q (LED_2)
0	0	hold	hold
1	0	1	0
0	1	0	1
1	1	forbidden	forbidden

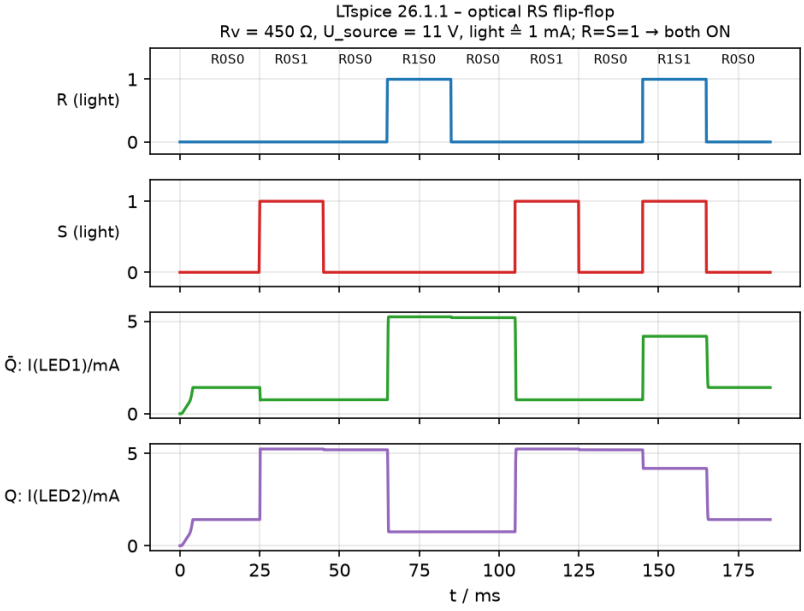


Fig. 6: LTspice – truth table with light pulses (R_V = 450 Ω, U_{source} = 11 V). With $R_V \approx 0$ both LEDs stay ON, so R_V is essential.

5. Pay attention during the lab

- Never exceed **15 V** source voltage.
- Change only **one** resistor at a time and write down all potentiometer positions.
- **Ambient light** (room light, phone torch) can switch the photodiode or LDR. Shade the box.

- Sweep the voltage **up and down**, otherwise there is no hysteresis to see. With $R_V > 0$, the x-axis is $\mathbf{U_{AB}}$, not U_{source}.
- Oscilloscope: **DC coupling**, both probe grounds on the **same node**, current via the voltage across a shunt (use INV if the sign is wrong).

6. Why do we need this later?

- **Optocouplers everywhere:** galvanic isolation in power electronics (gate drivers, PLC inputs, medical devices). The PC817 used here sits in the feedback loop of almost every switch-mode power supply (together with a TL431). CTR, speed and temperature drift are exactly the limits you meet there.
- **Schmitt triggers:** every robust digital input (74HC14), push-button debouncing, light barriers, sensor signal conditioning and simple oscillators.
- **Positive feedback and bistability:** latches, flip-flops and SRAM cells, the thyristor/TRIAC (same S-shaped curve), comparators with hysteresis, thermostats.
- **Load-line thinking:** transistor biasing, LED drivers, the maximum power point of solar cells, and the stability of negative-resistance devices (tunnel diode, Gunn diode, oscillators).
- **Photonics and optical computing:** optical switching in fibre networks (WDM, ROADM), spatial light modulators, optical neural networks and all-optical flip-flops. These are the research directions this hybrid circuit illustrates.
- **Lab skills:** X-Y measurements, quasi-static versus dynamic measurements, and the comparison of simulation and measurement. You will need these in every later lab and in your thesis.

7. Typical questions – short answers

- **Where does the hysteresis come from?** From positive feedback with a loop gain above 1.
- **Is the system linear? Time-invariant?** It is not linear, because superposition fails. It is time-invariant: having a memory does not make a system time-variant.
- **Why can the middle branch not be measured with $R_V = 0$?** Its negative differential resistance is unstable under voltage control. It needs current control, i.e. $R_V > |r_{NDR}|$.
- **What is the difference between the memory and the Schmitt trigger?** Only the operating point: bistable in the dark gives a memory, monostable in the dark gives a Schmitt trigger.
- **Why is R = S = 1 forbidden?** Both cells are forced ON ($Q = \bar{Q}$). When the light is released, the result is a race and the final state is undefined.
- **Phototransistor or photodiode?** The phototransistor has a high CTR (≈ 50–600 %) but is slow (µs). The photodiode has a CTR of only about 0.1–1 % but is fast (ns).